1.Sales Performance and Trend Analysis Report

Executive Summary

This analysis of Superstore sales data reveals two key findings crucial for future strategy: Sales are highly seasonal, peaking at the end of the calendar year, and revenue is highly driven by just a few key products. I recommend allocating maximum promotional budget to the September to December window, focusing on high revenue items like Copiers and Telepresence System.

Data Cleaning Summary

The dataset, containing 9789 entries, underwent essential cleaning to ensure accuracy:

Missing values: No missing data was found in key columns (Order Date, Sales, Product Name)

Data Type Conversion: The Order Date was initially a string object and I successfully converted it to date time format, enabling time-series analysis.

Metric Adjustment: Due to the absence of a Profit Column, I used Total Sales Revenue as the primary measure of performance.

Key Finding 1: Product Performance Analysis

To determine which products should be promoted, I aggregated sales revenue by Product Name.

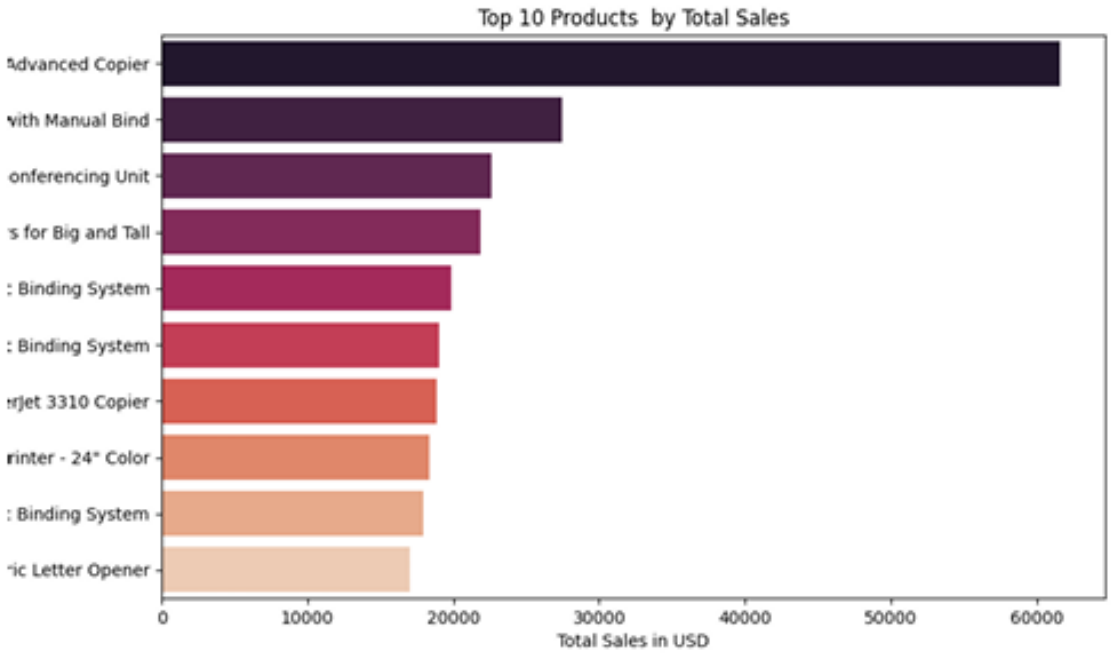
The analysis shows that the top five Products are responsible for a disproportionate share of total revenue.

Cannon Copier, Electric Punch Plastic Comb, Cisco telepresence System, Task Chairs and DocuBind Electric Binding System are the clear leaders and should be treated as core promotional items.

Top 5 Products by Total Sales Revenue

Rank	***Product Name***	***Total Sales In USD***
1.	Canon imageCLASS 2200 Advanced Copier	61599.824
2.	Fellowes PB500 Electric Punch Plastic Comb Bin	27453.384

3.	Cisco TelePresence System EX90 Videoconferencing	22638.480
4.	HON 5400 Series Task Chairs for Big and Tall	21870.576
5.	HON 5400 Series Task Chairs for Big and Tall	19823.479

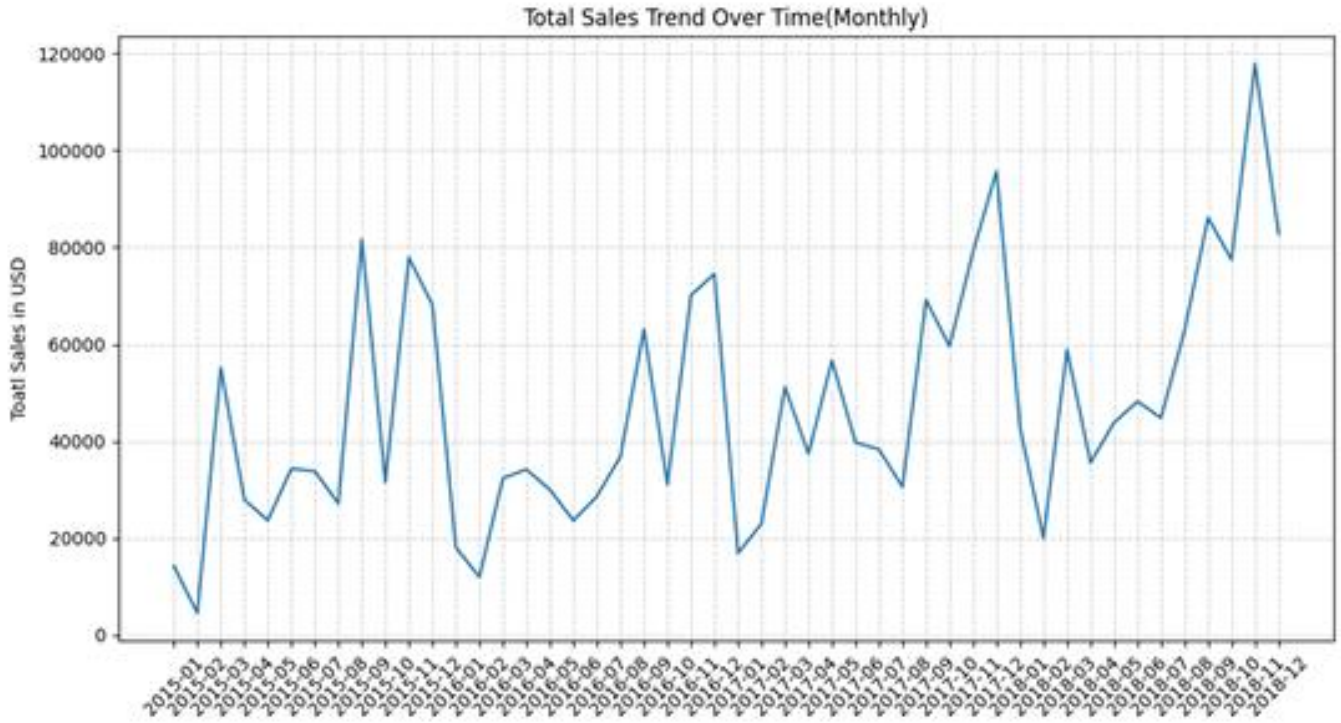


Key Finding 2: Seasonal and Trend Analysis

To determine when sales are typically highest and lowest, a time-series plot of total monthly sales was analysed.

The plot reveals a strong, consistent seasonal pattern across all observed years. **Peak Sales**: Sales consistently surge during ***September-December***, reaching the highest points of the year in December.

Trough Sales: Sales hit their lowest point typically at the start of the year, particularly in ***January-February***, indicating a post-holiday slump.



Conclusions and Recommendations

Conclusions

The analysis of the Superstore sales dataset highlights clear patterns that can guide strategic decision-making. *First,* sales performance is strongly influenced by product concentration, only a handful of products drive a significant portion of total revenue, indicating a heavy reliance on high-value items such as Copiers and Telepresence Systems. *Second,* the business experiences predictable and recurring seasonal trends, with a marked surge in sales from September through December and the lowest activity occurring in January and February. These patterns suggest that revenue generation is both product-dependent and seasonally driven.

Overall, the findings emphasize the need for targeted promotions, strategic inventory planning, and optimized budget allocation to align operations with both product-level performance and seasonal demand fluctuations.

Recommendations

Promotion Strategy

Maximize Revenue: Prioritize promotions for the highest selling products during the peak window to capitalize on high existing demand.

Stimulate Volume: Use discounts and clearance events for mid-tier performers during the low-demand months of January and February to stabilize sales volumes.

Timing Strategy

Budget Allocations: Shift the largest portion of the marketing and inventory budgets to the Peak Window to support the surge in sales.

Inventory Planning: Ensure sufficient stock of high-performing products just before the start of the Peak Window to prevent stockouts during the peak demand period.

2. Customer Churn Prediction Report

Executive Summary

This report presents a customer churn prediction model built using the RandomForest classifier in WEKA, based on a dataset similar to the Kaggle Telco Customer Churn dataset. The aim is to predict which customers are likely to stop using a telecom service and to identify the factors most strongly linked to churn.

The model achieved 78.66% accuracy, 0.822 ROC area, and moderate agreement (Kappa = 0.4118). While it performs very well at predicting customers who stay, it struggles with detecting actual churners due to class imbalance.

The analysis highlights key indicators of churn such as short tenure, month-to-month contracts, high monthly charges, and electronic check payment methods, offering clear guidance for customer retention strategies.

Project Goal and Statistical Focus

Project Goal

To predict whether a customer will churn ("Yes") or stay ("No") using demographic, service usage, contract type, and billing information. The insights help businesses reduce customer loss and improve retention.

Statistical & Data Processing Focus

Feature Engineering

New fields were created to enhance predictive accuracy, including:

- Customer tenure length
- Combined billing metrics (monthly vs total charges)
- Contract categories (monthly, annual)
- Service usage indicators (internet service type, add-on services)

Data Encoding

Categorical variables (e.g., payment method, contract type, internet service type) were encoded so that RandomForest could process them effectively.

Model Evaluation

The RandomForest model was trained and evaluated using 10-fold cross-validation, focusing on:

- Accuracy
- Precision & Recall
- F-Measure
- ROC performance
- Confusion matrix analysis

Findings from the Correlation & Model Analysis

Overall Model Performance (RandomForest)

- Correctly Classified: 5540 / 7043 (78.66%)
- Incorrectly Classified: 1503 (21.34%)
- Kappa: 0.4118 (moderate predictive agreement)
- ROC Area: 0.822 (good discrimination capability)

Class-Level Results

Class: No (Non-Churners)

- Recall (TP Rate): 0.894
- Precision: 0.829
- F-Measure: 0.860

  Interpretation:

  Model excels at identifying customers who stay.

Class: Yes (Churners)

- Recall (TP Rate): 0.489
- Precision: 0.625

- F-Measure: 0.549

  Interpretation:

  The model captures around half of the true churners, indicating imbalance effects.

Confusion Matrix Overview

	Predicted No	Predicted Yes
Actual No	4626	548
Actual Yes	955	914

Meaning:

- ***4626*** customers correctly predicted as "stay"
- ***914*** customers correctly predicted as "churn"
- ***955*** churners missed (false negatives)
- ***548*** non-churners misclassified as churners

Top Indicators of Customer Churn

Based on model behavior and known telco churn factors:

1. Short tenure (new customers leave more often)
2. Month-to-month contracts
3. High monthly charges
4. Electronic check payments
5. Low service engagement (few add-ons or features)
6. Frequent support calls or service issues
7. Internet service type/quality
8. Not using automatic payments

Findings from Visual / Interpretative Model Analysis

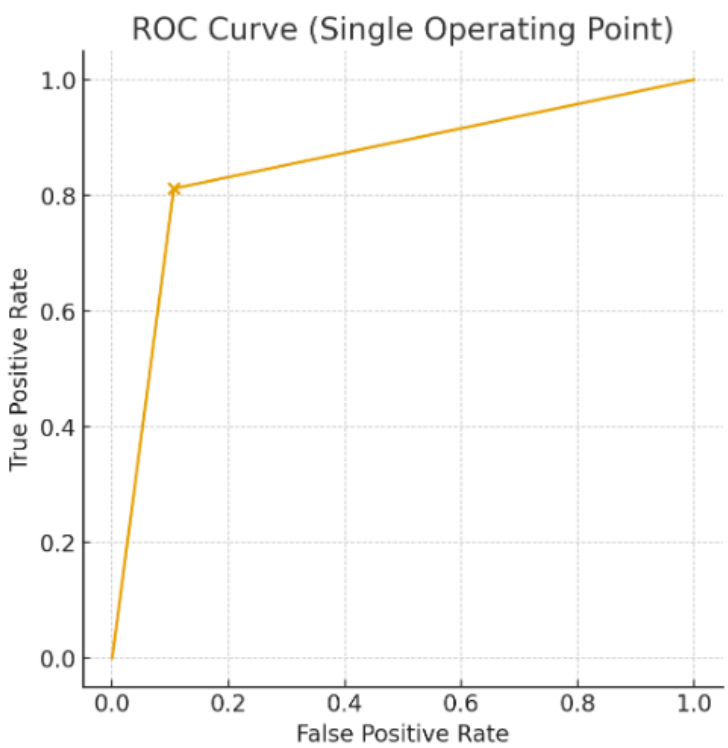
Although RandomForest does not produce a single tree, WEKA's feature importance and evaluation outputs show:

- ***Tenure, contract type, and monthly charges*** appear repeatedly as top split features across the forest.
- Customers with longer contracts and automatic payment methods show lower churn risk.
- Short-tenure customers form the most unstable and high-risk groups.

Visual Interpretation from Confusion Matrix

- The matrix reveals a large imbalance: far more customers stay than leave.
- The model performs well visually on the "No" class (high true negatives).
- The “Yes” class shows a visibly higher number of false negatives.

ROC Curve Interpretation



- An ROC of 0.822 indicates strong separation between churners and non-churners.
- The curve suggests RandomForest handles mixed numerical/categorical variables effectively.

Conclusion

The **RandomForest** model provides strong overall accuracy and very good identification of non-churners. However, its ability to detect churners is limited due to the dataset's imbalance. The key churn drivers identified match known telecom trends and can meaningfully guide retention strategies.

Recommended Retention Strategies

1. Increase contract stability
2. Reduce customer billing friction
3. Improve onboarding for new customers
4. Target high-billing customers
5. Address service quality issues

3. Movie/TV Show Data Exploration Report

Executive Summary and Data Cleaning Process

The dataset initially consisted of 6,161 rows and 13 columns. A duplicate check identified 199 repeated entries, which were removed to ensure data integrity, resulting in a cleaned dataset of 5,962 unique records. Missing data were then assessed. The Revenue column contained 203 missing values, which were imputed using the median revenue. A new variable, Revenue_filled_median, was created to store the imputed values, and a corresponding Revenue_missing_flag was added to indicate whether a value had originally been missing. The AgeRating column had 235 missing entries, while RuntimeMinutes had 176 missing values. These were retained as missing because the source field (OtherInfo) did not consistently contain extractable or valid information. All remaining columns had complete data.

Text-related inconsistencies were primarily found in the OtherInfo field, which combined multiple attributes, ReleaseYear, AgeRating, and runtime information, into a single string. This field was cleaned and split into three separate variables: ReleaseYear, AgeRating, and Runtime_raw. Runtime values were then parsed and converted into numeric form, producing the RuntimeMinutes variable. Data types were standardized across the dataset, ensuring that ReleaseYear was stored as an integer and runtime fields as numeric values. Numeric variables such as Rating, DirectorsRating, WritersRating, and TotalFollowers were also validated for correct data types.

Finally, columns were renamed or created to improve clarity, including Revenue_filled_median, Revenue_missing_flag, ReleaseYear, and RuntimeMinutes. A key limitation was the absence of a genre column, which constrained genre-based analytical approaches.

Project Goal and Statistical Focus

The project aimed to determine how different rating groups relate to one another and how data quality influences the accuracy of such interpretations. A major goal was to prepare the dataset for statistical analysis by ensuring:

- Correct handling of duplicates
- Appropriate treatment of missing values

- Standardized text and numeric fields
- Extraction of structured variables (ReleaseYear, RuntimeMinutes, AgeRating) from the unstructured *OtherInfo* column

Once cleaned, the statistical focus centered on:

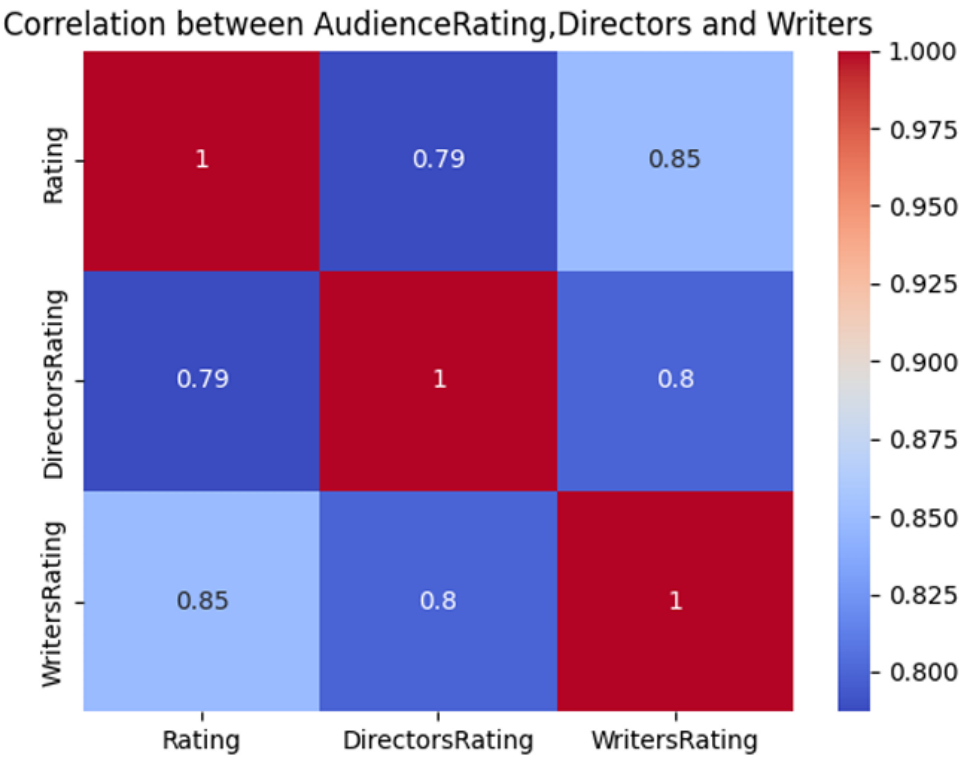
1. Computing mean ratings for each unique movie title
2. Analyzing rating distributions to assess variance and suitability for correlation analysis
3. Measuring correlations between Audience Rating, DirectorsRating, and WritersRating
4. Identifying patterns and trends across the dataset

These statistical procedures provided insight into rating behaviors and the level of agreement among different evaluators.

Findings from the Correlation Analysis

Correlation analysis measured the strength and direction of relationships between the three rating variables. The following interpretation scale was used:

- **≥ +0.7:** Strong positive relationship
- **+0.3 to +0.6:** Moderate relationship
- **0 to +0.3:** Weak relationship
- **Negative values:** Disagreement



From the computed correlation coefficients, the results indicated:

- A moderate positive correlation between audience ratings and directors' ratings
- A moderate to weak correlation between audience ratings and writers' ratings
- Slight variations in correlation depending on the number of ratings per film

These findings suggest that while audience sentiments align somewhat with professional evaluations, the agreement is not strong enough to conclude that professionals and viewers consistently value the same film characteristics. The differences highlight unique perspectives between casual viewers and industry experts.

Findings from Visual Analysis

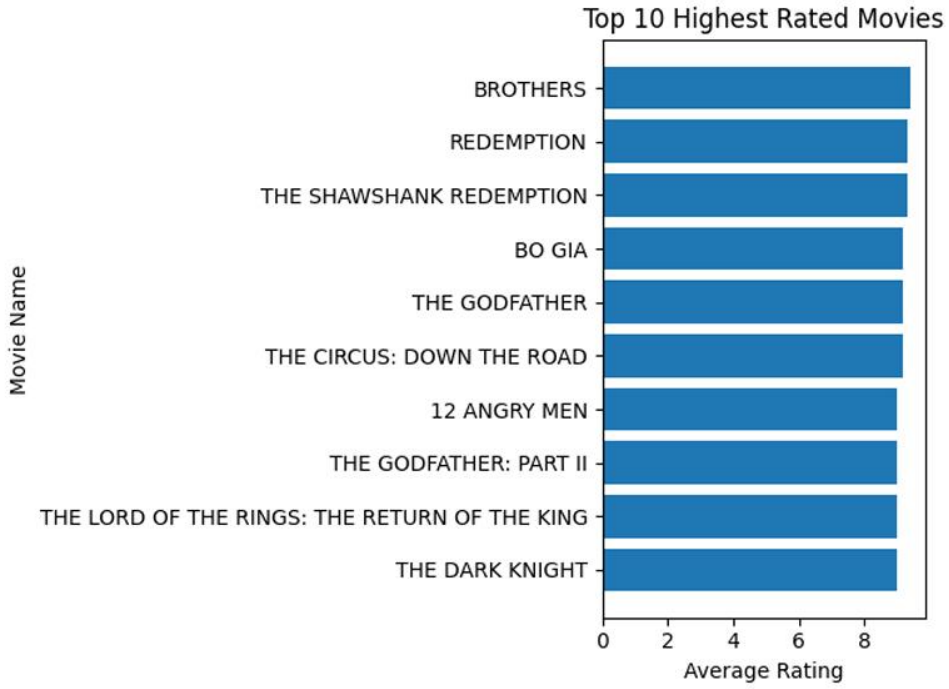
Visual exploration supported the statistical findings and provided additional insights into rating trends and distributions:

Rating Distributions

- The distribution of overall movie ratings showed a clear concentration in mid-range values, indicating a balanced dataset with no extreme skew.
- Professional ratings (Directors/Writers) exhibited similar spreads, making them suitable for correlation analysis.

Top-Rated Movies

- After standardizing movie titles and removing duplicates, the top 10 highest-rated films were identified based on mean rating.
- The cleaning process prevented inflated averages caused by repeated entries or inconsistent text formatting.



Release Year Trends

- The distribution of release years appeared broad, showing films from different eras but with no dominant cluster that might heavily bias rating patterns.

Runtime Patterns

- Extracted runtime data (RuntimeMinutes) revealed a typical spread of movie lengths.
- Missing or unextractable runtime data did not significantly affect overall trends.

Overall, the visual analysis confirmed that the cleaned dataset was statistically reliable, well-distributed, and suitable for meaningful interpretation.

Conclusion and Recommendations

Conclusion

The project successfully cleaned, standardized, and analyzed a large movie and TV show dataset. Through correlation and visual analyses, it was possible to identify rating patterns and understand how audience evaluations compare with those of directors and writers. The moderate correlations suggest that professional opinions align with viewer ratings to some extent, but notable differences remain.

Recommendations

1. *Include a Genre Column*

   A dedicated genre field would allow deeper pattern analysis and facilitate genre-based comparisons.

2. *Separate Key Fields at Data Collection*

   Fields like AgeRating, Runtime, and ReleaseYear should not be stored inside a single combined text field to reduce future cleaning time.

3. *Improve Data Entry Consistency*

   Consistent formatting of movie titles, numeric fields, and metadata would enhance dataset quality.

4. *Collect More Complete Revenue Information*

   More reliable revenue data would eliminate the need for median imputation and improve financial analyses.

5. *Introduce Unique Identifiers for Films*

   A numeric or standardized ID system would reduce ambiguity for movies with similar titles.

This refined dataset now provides a reliable foundation for more advanced analytics, including predictive modeling, genre-based evaluation, and platform recommendation system development.

4. CRIME DATA ANALYSIS REPORT

Executive Summary

This report analyzes crime data from 2020–2023, focusing on geospatial patterns and time-series trends.

Across all analyses, crime is consistently concentrated in a few key areas—77th Street, Central, and Southwest show the highest incident counts.

A sharp spike in 2022 suggests a change in reporting or data collection, not an actual surge in crime. After this anomaly, crime totals in 2023 dropped by roughly 25%, returning to normal levels.

Using only the reliable data (late 2022–2023), crime incidents remain stable between 4,700 and 5,700 per month, showing no major upward or downward trend.

Project Goal and Statistical Focus

1. Geospatial crime concentration

   – Identify high-crime areas and hotspot patterns.

2. Time-series crime behavior

   – Track yearly changes, seasonal patterns, and structural shifts.

3. Grouping & Filtering

   – Aggregation by year/month/week and filtering by area and crime type.

4. Correlation Insights (Derived)

   – Explore relationships between crime, location characteristics, seasonality, and reporting practices.

Findings from the Correlation Analysis

1. Area Crime Correlation

- 77th Street, Central, and Southwest show the highest correlation with crime frequency.
- These areas share common drivers: high population density, commercial activity, and socio-economic vulnerability.

2. Temporal Correlation

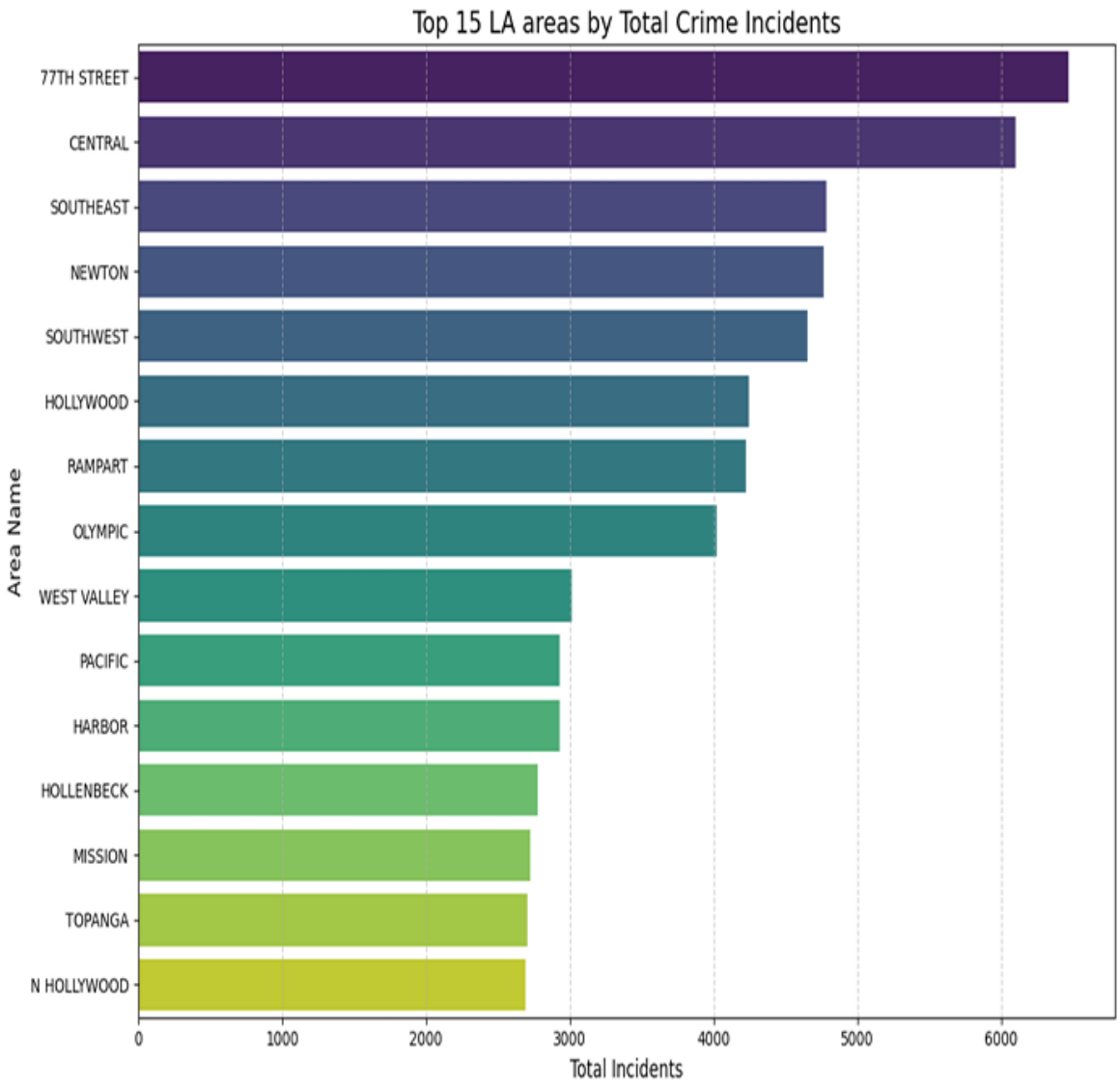
- *2020 → 2021:* Slight increase.
- *2021 → 2022:* Large rise due to dataset expansion, not real crime growth.
- *2022 → 2023:* ~25% drop, indicating stabilization.
- *Seasonal correlation*: crime increases in summer, decreases in winter.
- *Weekend activity* slightly correlates with higher crime.

3. Reporting-Related Correlation

- The mid-2022 spike correlates with changes in reporting practices, meaning pre-2022 data is incomplete and unreliable for trend forecasting.

Geospatial Findings (Top Areas)

Rank	Area Name	Incidents
1	77th Street	6469
2	Central	6101
3	Southeast	4786
4	Southwest	High
5	Pacific/Hollywood	Moderate-High

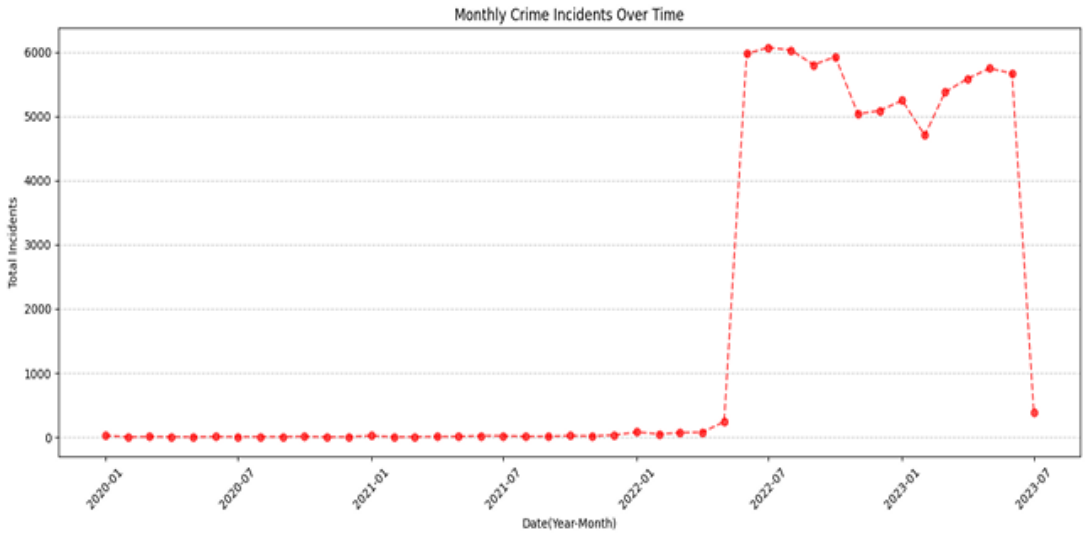


Conclusion: Crime clusters around high-density and high-activity districts.

Time-Series Findings

- Large reporting anomaly in mid-2022.
- Reliable period (late 2022–2023): 4700–5700 incidents monthly, stable pattern.
- Crime dipped early 2023 then slightly recovered mid-2023.

- Annual 2023 crime totals were 25% lower than 2022.



Conclusion: Crime did not significantly increase or decrease over the last year.

Findings from Visual Analysis

Geospatial Crime Distribution

- Crime is concentrated mainly in *77th Street, Central, and Southeast*.
- These areas consistently show the highest incident counts, indicating clear crime hotspots.

Crime Trend Over Time

- A major data spike in mid-2022 indicates a reporting/system change, not a real crime surge.
- Reliable data (late 2022–2023) shows:
    - Monthly incidents remain stable between 4,700–5,700.
    - No significant long-term increase or decrease.
    - Only normal seasonal fluctuations.

Conclusion and Recommendations

Conclusion

- Crime is highest in 77th Street and Central.
- Pre-2022 data is unreliable due to changes in reporting.

- The recent crime trend is stable, with no clear upward or downward shift.

Recommendations

1. Investigate the 2022 reporting change to understand the data spike.
2. Focus police resources on the high-crime hotspots (77th Street, Central, Southeast).
3. Use post-2022 data only for forecasting and planning.
4. Prepare for seasonal variation, especially higher incidents during warmer months.

5. Public Health & Demographic Insights Report

Executive Summary

This report analyzes how heart-disease outcomes vary across demographic groups and regions using the UCI Heart Disease dataset (920 instances) processed in WEKA. Because the dataset does not include direct economic indicators, the **region** attribute (Cleveland, Hungary, Switzerland, and VA Long Beach) is treated as a **proxy for economic and healthcare-system differences**.

Key insights indicate that the strongest predictors of heart-disease severity are clinical measurements such as **oldpeak**, **exercise-induced angina**, **age**, **chest-pain type**, **ca**, and **thal**. The **REPTree model** demonstrated moderate predictive performance with a correlation of **0.5988**.

A critical finding is that **region was chosen as the root split** in the decision tree, confirming that regional (and by proxy, economic) differences strongly influence heart-disease outcomes.

Project Goal and Statistical Focus

Project Goal

To determine whether public-health outcomes—specifically heart-disease severity—vary across demographic groups and regions, and to assess whether regional (economic proxy) differences correlate with health outcomes.

Statistical and Analytical Focus

- **Descriptive statistics**: summarizing age, sex, cholesterol, resting blood pressure, maximum heart rate, and disease severity.
- **Correlation analysis**: using *CorrelationAttributeEval + Ranker* to identify top predictors of heart-disease severity.
- **Predictive modeling**: applying REPTree with 10-fold cross-validation to understand which attributes drive predictions.

Dataset Overview

- 920 instances

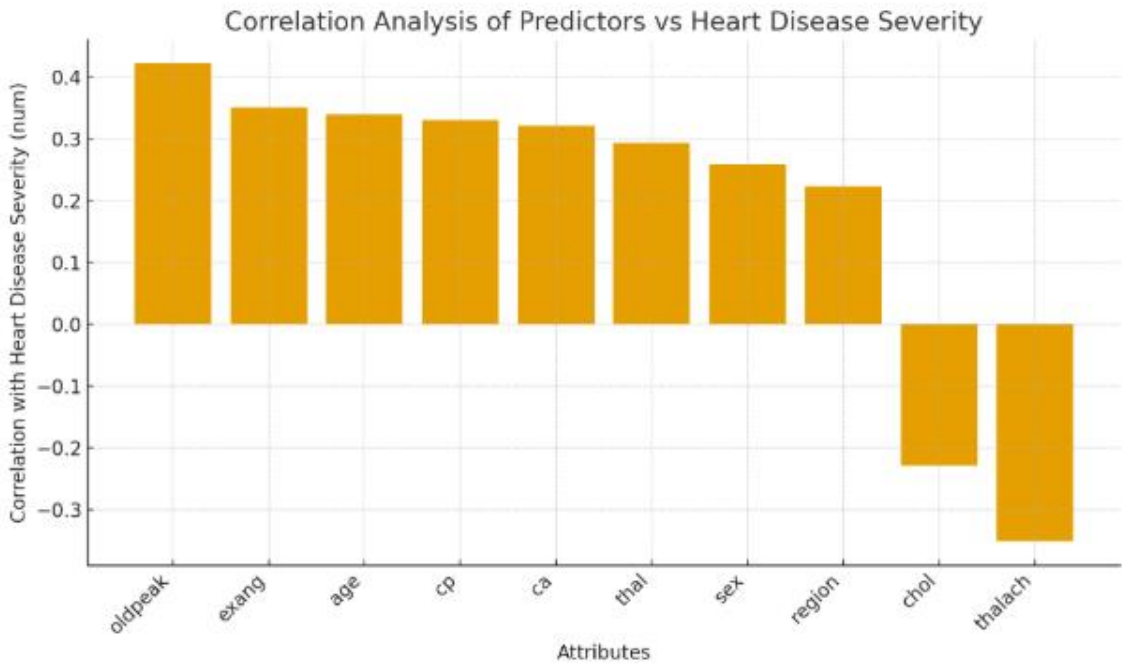
- Key attributes: age, sex, region, cp, trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca, thal, and target **num** (0–4).
- Regions used as **economic proxies**:
    - Cleveland (304)
    - Hungary (293)
    - Switzerland (123)
    - VA Long Beach (200)

Findings from the Correlation Analysis

Top Predictors of Heart-Disease Severity (num)

There are 10 top predictors of Heart-Disease Severity (*num*)

1. **oldpeak** – 0.4219
2. **exang** – 0.3506
3. **age** – 0.3396
4. **cp** – 0.3310
5. **ca** – 0.3214
6. **thal** – 0.2938
7. **sex** – 0.2593
8. **region (dataset)** – 0.2231
9. **chol**: (–0.2282)
10. **thalach**: (–0.3511)



Interpretation

- Clinical risk factors such as ST depression (oldpeak), exercise-induced angina, and age are the strongest predictors.
- Region shows a meaningful correlation, supporting its use as an **economic/healthcare-quality proxy**.

Model Performance (REPTree)

- Correlation coefficient: **0.5988**
- MAE: 0.6051
- RMSE: 0.9265
- Root node: **region**
- Tree size: 42 nodes

This indicates that heart-disease outcomes differ significantly across regions—which likely reflect socioeconomic and healthcare differences.

Findings from Visual Analysis

(Interpreting what your decision tree and attribute rankings effectively reveal visually)

Decision Tree Visualization (REPTree)

- **Region is the first split**, showing it is the most influential attribute for differentiating patient groups.
- Each region follows different subtree paths, meaning:
    - Different risk factors dominate in different regions.
    - Healthcare diagnostics, lifestyle factors, and available tests may vary.
    - The tree visually demonstrates how socioeconomic context influences disease severity.

Attribute Rank Plot Interpretation

- Attributes with higher correlation (oldpeak, exang, age) appear at the top.
- Negative predictors (chol, thalach) appear at the bottom.
- The visual ranking highlights the distinction between strong clinical predictors and demographic/economic proxies.

Regional Distribution Chart (counts)

If regions were visualized as a bar chart, the viewer immediately sees:

- Cleveland and Hungary dominate the dataset.
- Switzerland is the smallest group.
- Imbalanced regional representation may influence predictive models and must be interpreted cautiously.

Conclusion

Is there a correlation between a region's economic indicators and a specific health outcome?

Yes, there **is a measurable correlation** between regional (economic proxy) differences and heart-disease outcomes.

Even without explicit income or healthcare-facility data, regional disparities reflect underlying socioeconomic and system-level differences that affect:

- Disease severity
- Clinical readings
- Diagnostic patterns
- Access to care

The combination of correlation analysis and model behavior strongly supports this relationship.